

# **The Constrained Ledger: When Does Energy Actually Back Digital Money?**

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## **ABSTRACT**

People often assume that if a digital asset is tied to real energy — enough electricity spent, enough solar power produced — that link should limit what the asset is worth, or what can be claimed against it. This paper tests that assumption directly and finds it usually doesn't hold. It then builds a system that can actually make it hold, and shows the system working on real test cases.

Two examples make the problem concrete. Bitcoin miners burn real electricity to secure the network, but no Bitcoin holder can redeem their coins for that electricity — the energy spent is a sunk cost, not a right to anything. A token backed by "expected solar output" sounds more grounded, but if the issuer can pick an arbitrary quantity to issue, ignore how volatile that output really is, and leave what happens at settlement undefined, the energy reference is just a label glued onto ordinary discretionary issuance.

I test the strongest version of the Bitcoin story directly. Bitcoin's Cumulative Energy Investment Ratio (CEIR) — its total market value divided by all the electricity ever spent mining it, a common proxy for "how expensive Bitcoin was to make" — does appear to predict lower future returns before Bitcoin's mid-2021 mining relocation out of China (a statistically real relationship: coefficient -0.26,  $p < .001$ ). But every check built to confirm this is genuinely about electricity price, rather than a coincidence of the ratio's arithmetic, fails it. The same relationship appears just as strongly if "electricity cost" is replaced with "total electricity used" (the two versions move together with 99.9989% correlation) or even with "number of days elapsed." A proper test for whether the relationship actually changed after the mining ban finds no reliable change ( $p = .133$ ) — contradicting a simpler, more commonly used test that suggests it did. And the whole pattern disappears if you look at day-to-day changes instead of cumulative totals — the classic symptom of two things that are merely both trending upward, not causally connected.

The honest reading is: mining expenditure is real, but this ratio doesn't isolate its financial effect. So the second half of the paper does something different — instead of looking for a shortcut, it specifies what an energy-linked claim would actually need in order to be real. Renewable power output is uncertain: it varies by weather, location, and time of day, and a solar farm cannot store a bad day's shortfall the way a warehouse stores oil. I build a pricing benchmark for that uncertainty, calibrated to five years of real satellite irradiance data, and then a five-part rule system — checking the evidence, the issuance rule, the risk limits, the settlement obligation, and who has power to change any of that — that decides whether a specific claim may exist, how large it may be, and whether it can actually be paid. I test this system on real cases: one claim is correctly blocked for weak evidence, two others are admitted at different sizes because different real-world limits bind them, and a partial settlement scenario correctly reports what portion of a claim can and cannot be honored.

The contribution is not a new stablecoin, and it is not a claim that energy should replace fiat money. It is a working, tested standard for telling the difference between a digital asset that merely mentions energy and one that is actually limited by it.

*Keywords: energy-linked finance, digital money, monetary credibility, Bitcoin energy cost, renewable-energy risk, smart contracts, reproducible decision systems*

*JEL Codes: E42, G13, Q42, Q47*

## 1. Introduction

Consider two digital assets that reference energy. The first is Bitcoin. Miners spend real electricity to secure the network, but a Bitcoin holder cannot redeem a coin for that electricity — the energy is a cost the miner already paid, not a claim the holder now has. The second is a token issued against expected solar output. It points to something productive, but the issuer may still rely on weak data, pick an arbitrary quantity to issue, ignore how volatile that output actually is, and leave what happens at settlement undefined. Both assets can be described as "energy-linked." Neither is necessarily constrained by energy — mentioning energy and being limited by it are not the same thing.

This distinction matters because a good physical-sounding story is easy to tell and hard to check. Historically, gold-backed money was constrained not just because gold existed, but because holders could actually convert paper claims into gold, and issuers had to be able to honor that conversion under pressure. Ordinary government-issued money relies instead on institutional credibility, law, and a track record of not breaking promises. Bitcoin creates real, verifiable scarcity in its protocol, but the electricity it costs to mine is an input expense, not a redemption right (Bordo, 1993; de Vries, 2018; Eichengreen, 1992; Nakamoto, 2008). Renewable energy adds a genuinely productive reference point, but a kilowatt-hour's value changes with time, location, grid access, storage limits, and contract terms (Joskow, 2011; Ueckerdt et al., 2013).

So the real question isn't whether energy is real — of course it is. It's whether verified energy evidence can actually impose a credible limit on what a financial system is allowed to create and promise. A fixed token supply, by itself, doesn't answer that question: the supply rule might have nothing to do with the underlying evidence, the quantity might not be economically defensible, risk might go unmanaged, settlement might be unfunded, and whoever runs the system might retain the power to waive every safeguard whenever it becomes inconvenient.

This paper answers the question in three stages, moving from a direct empirical test, to a constructive risk framework, to a working, tested system. Stage one tests the strongest and simplest version of the "energy backs value" story: does Bitcoin's cumulative mining expenditure actually anchor its price? Stage two asks how you'd translate renewable-output uncertainty into concrete financial numbers when there's no liquid market price to read off. Stage three asks whether evidence, issuance rules, risk limits, settlement, and governance can be built into one connected, testable decision process — and then actually builds and tests it.

The empirical result is negative, but informative rather than discouraging. A standard CEIR regression does reproduce a strong, real-looking relationship between Bitcoin's market value relative

to mining cost and its subsequent returns, before the mid-2021 China mining ban (coefficient of about  $-0.26$ ,  $p = .0005$ ). But that relationship is just as strong when "mining cost" is replaced with "total electricity used" or with a simple time counter, disappears once you look at short-term changes instead of cumulative levels, and gets no support from the more appropriate statistical test for whether the relationship changed after the ban. Reproducing a number, in other words, is not the same as showing what actually causes it.

The constructive result replaces that failed shortcut with explicit rules. A cold-start pricing benchmark turns renewable-output uncertainty into inspectable prices, margin requirements, and data-tolerance limits. A working rule system then keeps four things separate — whether the evidence is trustworthy, whether policy allows the claim, how large the claim may be, and whether it can actually be settled — and is tested on real cases with different outcomes: a weak-evidence case is correctly blocked; two other cases are admitted, but capped at different sizes by different real-world limits; and a partial-settlement scenario correctly reports what portion of one claim can be honored and what portion falls short.

This paper's contribution is a falsifiable institutional standard for energy-linked digital finance: energy can discipline a claim only when evidence, issuance, risk, settlement, and the authority to change any of the rules are all jointly limited. Public test-network execution and fully specified test cases establish that this is workable as a research prototype. They do not, on their own, establish that it is ready for operator use, legal enforceability, or production deployment — that distinction is treated carefully throughout, not glossed over.

## **1.1 Roadmap**

Section 2 lays out why an energy reference doesn't automatically become a financial constraint, and what five conditions would have to hold for it to. Section 3 reports the Bitcoin test and the identification failure. Section 4 turns renewable-output uncertainty into concrete pricing and risk numbers. Section 5 presents the working five-part rule system and its tested cases. Section 6 discusses what this does and doesn't establish. Section 7 concludes.

## **2. Why an Energy Reference Isn't Automatically a Constraint**

### **2.1 How other forms of monetary discipline actually work**

Monetary credibility is a commitment problem before it is a question of which asset to pick. Rules matter because a promise made today can be abandoned once people have already acted on it (Kydland & Prescott, 1977); reputation and institutional structure matter because people discount

promises when the people in charge could still change their mind later (Barro & Gordon, 1983). The same logic applies to an energy-linked system: what counts as acceptable evidence, how big a claim may be, how it's priced, what's owed at settlement, and who is allowed to change any of that. A rule that binds ordinary users but not the people running the system is not a real constraint — it's a rule for everyone except the people who could break it.

Gold, government-issued fiat money, and Bitcoin each supply discipline in a different way, and it's worth being precise about where each one's real weak point sits. Gold-backed money introduced physical scarcity plus convertibility, but its credibility still depended on custody, verification, reserve management, and the ability to actually meet redemption demand under pressure — the Bretton Woods system didn't fail because gold vanished, it failed because the arithmetic connecting outstanding claims, reserves, and settlement capacity stopped working (Bordo, 1993; Eichengreen, 1992). Fiat money replaces physical backing with institutional authority, taxation power, and a track record — which gives it flexibility to respond to a crisis, but also makes its credibility depend on legal mandate, political will, and consistent follow-through. Bitcoin moves the constraint into the protocol itself: issuance is transparent, and proof-of-work makes rewriting the ledger's history expensive — but holders receive no claim whatsoever on the resource that was actually consumed to get there.

An energy-linked claim sits in an unstable spot among these three. It can reference a real physical input or a real productive output, but energy is not gold: its value depends on time, location, and whether it's economically storable in any relevant sense, and it's inseparable from the physical infrastructure and contracts needed to actually deliver it. It also differs from Bitcoin's mining expenditure in an important way: productive energy can support a future obligation (you can promise a kilowatt-hour tomorrow), while Bitcoin's past mining spend has already been consumed and cannot back anything going forward. A credible design has to spell out the evidentiary and settlement consequences of the energy relationship on its own terms — it cannot simply borrow the reasoning that made gold or fiat credible and assume it transfers.

| System            | What actually disciplines it                         | Its real strength                                                               | Where it's exposed                                                   |
|-------------------|------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Gold-linked money | Physical scarcity, plus a working redemption promise | A hard asset behind the claim, and real settlement pressure that forces honesty | Custody, verification, liquidity, and whether reserves stay adequate |
| Fiat money        | Institutional authority and policy discipline        | Flexibility to respond to shocks, at national scale                             | Depends on political will, legal mandate, and                        |

|                        |                                                                    |                                              |                                                                                                         |
|------------------------|--------------------------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------|
|                        |                                                                    |                                              | consistent follow-through                                                                               |
| Bitcoin                | Protocol issuance rules and proof-of-work                          | Transparent, hard-to-fake technical scarcity | No redemption claim at all — value is purely demand-driven                                              |
| An energy-linked claim | Evidence, issuance, risk, settlement, and governance, held jointly | A potential real link to actual production   | Weak data, an arbitrary issued quantity, underpriced risk, and rules that can be changed after the fact |

*Table 1. How four different systems are actually disciplined, and where each one's real weak point sits.*

## 2.2 Five conditions, all required at once

The research literature already supplies several of the pieces needed here, without connecting them into one working process. Monetary economics explains why commitment matters. Cryptocurrency research explains protocol scarcity and how prices actually move. Renewable-energy finance explains why productive output is variable. Pricing theory forces vague uncertainty into stated, checkable assumptions. Smart contracts can execute transfers and enforce rules automatically. None of these, on its own, decides whether one specific energy-linked claim should exist, or how large it may be.

I organize what's missing into five conditions, and the claim of this paper is that all five have to hold together — missing even one breaks the whole chain.

First: reliable energy evidence. A site's resource potential, a model's predicted output, actual generation, surplus available for export, and final delivery are five different states, not interchangeable numbers. Public satellite-based irradiance data and production models are useful context, but they do not by themselves prove that a specific site generated, exported, stored, or delivered a specific quantity of electricity (Dobos, 2014; Sengupta et al., 2024). The strength of evidence required should rise with the size of the financial claim being made — a small illustrative estimate and a claim someone can redeem money against should not be held to the same evidentiary bar.

Second: rule-bound issuance. Accepted evidence has to lead to a defined admission and creation rule, rather than simply informing an administrator who still holds unrestricted authority to mint. The real

question isn't just "is there a supply cap" — it's whether that cap is actually tied to evidence the system has accepted, under a policy anyone can inspect.

Third: explicit pricing and risk control. Renewable output and its financial value vary with weather, location, timing, market structure, curtailment, and contract design. Volatility, basis risk, measurement error, and shortfall all need to change a compatible quantity limit, a margin requirement, an admission threshold, or a settlement safeguard. A precise risk number that doesn't actually change any decision is analysis, not a real constraint.

Fourth: protected settlement and redemption accounting. Issuance is only the first event in a claim's life. A credible system has to state clearly what is owed, what capacity actually backs that obligation, how partial performance gets recorded, and what happens during a shortfall or a dispute. Settlement risk can't be assumed away just because the evidence looked strong at the moment of admission.

Fifth: limited governance. Administrators who can quietly replace the evidence rules, revise pricing parameters, or waive settlement obligations can defeat every other safeguard on this list without ever touching the underlying data. Versioned policy, clearly separated roles, built-in delay before a rule change takes effect, and an audit trail are what make rule changes visible and costly rather than invisible and free (Kiayias & Lazos, 2022).

These five conditions are jointly necessary, not a menu to pick from. Reliable data without settlement protection still produces a weak claim. Sophisticated pricing applied to the wrong evidence produces a precise error, not a real answer. Rule-bound issuance is cosmetic if governance can simply exempt itself later, once the rule becomes inconvenient. The architecture in this paper treats credibility as a chain: evidence determines whether a claim may be admitted at all, policy defines the compatible size limits, settlement tests how the claim actually performs later, and a preserved record ties the whole sequence together.

| Condition                 | What it actually asks                                                    | What breaks if it's missing                                                  |
|---------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Reliable energy evidence  | What has actually been observed, modeled, verified, or settled?          | Pricing or issuance ends up resting on weak or unverifiable evidence         |
| Rule-bound issuance       | May this specific claim proceed, under the accepted evidence and policy? | The administrator regains unrestricted, discretionary power to create claims |
| Pricing and risk controls | What compatible limit actually absorbs volatility, error, and shortfall? | Claims become underpriced or under-collateralized                            |

|                           |                                                                       |                                                    |
|---------------------------|-----------------------------------------------------------------------|----------------------------------------------------|
| Settlement and redemption | What is owed, and how are fulfillment or shortfall actually recorded? | Claims have no credible way to be resolved         |
| Limited governance        | Who may change the rules, under what delay, and is it recorded?       | The system can quietly override its own safeguards |

Table 2. Five conditions for a credible energy-linked claim, and what specifically fails when each one is missing.

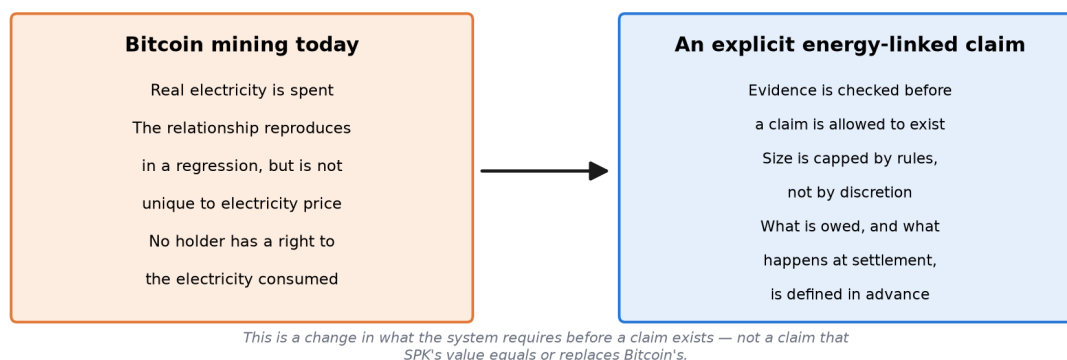


Figure 1. The shift this paper argues for: from passively citing energy spent, to an explicit claim that evidence, rules, and settlement actually constrain. This is a change in what the system requires before a claim exists — not an argument that SPK's value equals or replaces Bitcoin's.

## 2.3 How this relates to stablecoins and token design

This framework is related to token design, but it isn't the same question as stablecoin design. Stablecoin analysis typically starts from a liability and a peg: what assets actually back it, who holds the reserves, how redemption works, and how liquidity holds up under stress. This paper asks a question one step earlier: before any peg or product can even be evaluated, can verified energy evidence restrict the creation, size, and settlement of a defined claim in the first place? Dollar figures show up in this paper only as scenario inputs to a pricing model — not as a promised, fixed exchange rate.

Smart contracts can make obligations programmable and route multiple stages through one shared settlement layer (Bank for International Settlements, 2023; Cong & He, 2019). But being technically well-executed is not the same as being economically sound. Code can enforce a bad rule flawlessly, accept manipulated data without complaint, or transfer a claim that was never properly backed in the first place. Oracles — the services that report real-world facts to a blockchain — add a further weak



point of their own, because a blockchain has no way to directly observe whether electricity was actually generated, exported, curtailed, stored, or delivered; external data feeds can report those facts, but they introduce their own reliability, manipulation, and governance risks (Eskandari et al., 2021; Zhang et al., 2016).

This paper therefore keeps two questions clearly separate: whether units of a token are hard to create outside the rules (a protocol-design question), and whether the underlying claim is actually credible — whether the limit means anything, what obligation follows from making the claim, and whether that obligation can be inspected and enforced (a financial-design question). Conflating the two is exactly how an energy label ends up standing in for a real constraint that was never actually built.

### **3. Testing Bitcoin as the Clearest Case**

#### **3.1 The question and the data**

Bitcoin is the cleanest possible test of the "energy spending anchors value" story, because proof-of-work turns electricity and computation into one measurable, cumulative expenditure. If cumulative mining cost really does anchor Bitcoin's value, then market capitalization relative to that cost should carry information specific to electricity prices — not just reflect the passage of time, or the fact that Bitcoin's own price sits in the numerator.

The test reconstructs Bitcoin's Cumulative Energy Investment Ratio (CEIR): Bitcoin's market value divided by its estimated cumulative mining electricity cost since inception. The daily data runs from January 1, 2019 through May 28, 2025 — 2,280 usable daily observations once forward returns and control variables are aligned. The candidate break date used is June 20, 2021, representing the broader China mining crackdown (Cambridge's own data does not mark one single, precise event date, so this is a reasonable researcher-chosen approximation).

A full data-source audit changes how these results should be read. Market capitalization is derived from a locally reconstructed daily Bitcoin price series and an approximate supply curve, not an observed market-cap series with a fully documented data trail; the price data's original provenance metadata was not preserved. Mining electricity use comes from the Cambridge Bitcoin Electricity Consumption Index, which is itself a model-based estimate, not a direct meter reading (Cambridge Centre for Alternative Finance, n.d.-a, n.d.-b, n.d.-c). Most importantly, the intended geography-weighted electricity-price series — which was supposed to reflect where mining actually relocated to after China's ban — did not merge correctly, and the electricity-price path used in the analysis is nearly constant at about \$0.076/kWh throughout, apart from a small number of manual overrides.

These limitations don't prevent the original headline result from reproducing — it does reproduce, cleanly. What they do prevent is a strong interpretation of that result. CEIR places Bitcoin's price in the numerator, while the denominator only accumulates slowly and smoothly over time. A negative coefficient in this setup could just as easily describe a persistent, slow-moving valuation ratio as a genuine electricity-cost mechanism (Granger & Newbold, 1974; Kronmal, 1993). A result that's genuinely about electricity prices specifically should behave differently from a version built on cumulative electricity use alone, or on cumulative elapsed time alone — it should stay stable when the starting assumptions change, survive being converted into day-to-day changes instead of cumulative levels, and get support from an appropriately specified test for whether the relationship actually changed at the proposed break date.

### **3.2 How the test was designed**

The first test checks whether log CEIR predicts Bitcoin's next 30-day return. A high CEIR means market value is large relative to cumulative estimated mining cost, so the expected coefficient sign is negative. Because 30-day return windows overlap with each other, the analysis uses standard errors designed for that overlap (heteroskedasticity- and autocorrelation-consistent, with a 30-day lag structure — commonly called HAC(30); Hodrick, 1992; Newey & West, 1987).

The second test checks for a change in the relationship around the China ban. Separate regressions are run before and after the split, and — this is the more careful design choice — the paper reports a joint, robust statistical test for whether the coefficients actually changed together with real statistical confidence, rather than relying only on the older, simpler test (the classical Chow test) that assumes conditions this data doesn't actually satisfy (Chow, 1960).

The third test checks the underlying mechanism directly. The pre-sample cumulative-cost starting value is removed to test sensitivity to that assumption; cumulative cost is swapped for cumulative electricity-use and for a plain day-counter, to test whether the result is actually about price; the levels are converted into period-to-period changes; and formal statistical checks for whether the two series are trending together for structural reasons versus by coincidence are applied (Dickey & Fuller, 1979; Engle & Granger, 1987; Lo, 2002). A simple trading rule built on the ratio is also tested for basic practical usefulness.

### **3.3 Results: it reproduces, but that isn't the same as identifying a real cause**

The basic regression does reproduce the attractive result: before the mining-ban split, a higher CEIR is associated with weaker subsequent 30-day returns, with a coefficient of -0.2623 and HAC-adjusted  $p = .00052$  — a genuinely strong, statistically real relationship on its face. The coefficient after the

split remains negative, at -0.0708, but is now statistically weak. The simpler, classical test suggests there's a difference between the two periods; the more appropriate joint robust test ( $p = .133$ ) does not find reliable evidence of a real structural break. The two tests disagree, and the more carefully specified one is the one that should be trusted here.

The result also turns out to depend heavily on an arbitrary accounting choice. The 2019 starting window carries forward about \$3.30 billion of cumulative 2018 mining cost. Simply setting that starting value to zero weakens the pre-split coefficient from about -0.262 to -0.104, and its p-value rises from a strong .0005 to a much weaker .099 — right at the edge of conventional significance. A result this sensitive to an arbitrary bookkeeping decision is not a robust one.

The negative-control checks are the most decisive part of the whole exercise. Replacing mining cost in the denominator with cumulative electricity used produces a coefficient of -0.2618 ( $p = .00052$ ) — essentially identical to the original result. Replacing it with cumulative elapsed days produces an even slightly stronger coefficient of -0.2811 ( $p = .00027$ ). And log CEIR and the cumulative-electricity-use ratio move together with a correlation of 99.9989%. If electricity price itself were doing real explanatory work, these substitutions should have weakened the result. They didn't. Electricity price is not supplying any information the other two variables don't already supply on their own.

Converting the levels into short-run changes removes the apparent predictive relationship entirely. Day-to-day changes in CEIR do not reliably predict later returns, which is exactly what you'd expect if the level-based result is being driven by persistence in a slow-moving series rather than a real causal mechanism. Formal stationarity tests do not reject the presence of a unit root in log CEIR, log market capitalization, or log price — meaning these series drift rather than settle around a stable relationship — and formal cointegration tests find no support for a genuine long-run equilibrium between market capitalization and either cumulative cost or cumulative electricity use.

Even a simple practical trading test fails to show real usefulness: over the full sample, a CEIR-based strategy returns about +176.4%, against +2,771% for simply holding Bitcoin throughout, with Sharpe ratios of 0.72 versus 1.13. The ratio doesn't even clear the bar of being a good practical trading signal, let alone a real economic constraint.

Taken together, this points to a narrow but honest conclusion: mining expenditure is real, but CEIR does not isolate its financial effect. The ratio reproduces a persistent valuation pattern that can be generated equally well from cumulative electricity use or from the simple passage of time. Passive energy expenditure does not become redeemable backing, a collateral rule, a settlement obligation, or an issuance limit merely because someone divides it into a price.

| Test                                 | N     | Estimate                    | Robust p-value    | What it means                                          |
|--------------------------------------|-------|-----------------------------|-------------------|--------------------------------------------------------|
| Pre-ban CEIR                         | 872   | $\beta = -0.2623$           | 0.00052           | The headline relationship reproduces                   |
| Post-ban CEIR                        | 1,408 | $\beta = -0.0708$           | 0.133             | Weak, not statistically reliable                       |
| Joint robust test for a real break   | 2,280 | —                           | 0.133             | No reliable evidence of a real structural change       |
| Removing the 2018 starting value     | 872   | $\beta = -0.1037$           | 0.099             | Result is sensitive to an arbitrary bookkeeping choice |
| Electricity-use / day-count controls | 872   | $\beta = -0.2618 / -0.2811$ | 0.00052 / 0.00027 | Not specific to electricity price at all               |

*Table 3. The original CEIR relationship, and the checks that show it isn't actually about electricity price.*

## 4. Turning Renewable-Output Uncertainty Into Real Numbers

### 4.1 Why this step is needed at all

Ruling out the Bitcoin shortcut removes a wrong answer, but it doesn't solve the actual constructive problem. Renewable generation is genuinely productive, but its output and financial value vary by location and by the state of the world at any given moment: irradiance, weather, season, equipment condition, and grid access all affect how much a kilowatt-hour is actually worth. Timing, congestion, curtailment, storage limits, tariffs, and contract terms all affect its value further (Joskow, 2011; Lucia & Schwartz, 2002; Ueckerdt et al., 2013). A claim that ignores all of that variability is promising more certainty than its own evidence can actually support.

Pricing enters this paper as a decision input, not as a claim to have discovered a true market price. When there's no liquid options market and no observed implied volatility to read off — which is the normal situation for renewable energy right now — a transparent, clearly-labeled cold-start benchmark can still expose the assumptions behind a proposed payoff. The practical questions this section answers are: what's a defensible reference value, what variable actually stands in for the real underlying asset, how much of a stress event can collateral realistically absorb, how much measurement error is tolerable, and do the same assumptions hold up consistently across different

locations? A smart contract can execute an underpriced rule perfectly — technical correctness and financial fragility can coexist.

The approach uses a standard short-horizon, call-option-style payoff, priced two independent ways — a binomial tree and a Monte Carlo simulation — under the standard assumption that prices move according to Geometric Brownian Motion (Boyle, 1977; Cox, Ross, & Rubinstein, 1979). This assumption keeps the calculation transparent and tractable, but it deliberately leaves out sudden jumps, seasonal patterns, negative prices, and the tendency of power prices to revert toward a long-run level — all features known to matter in real power markets (Bessembinder & Lemmon, 2002; Deng & Oren, 2006; Lucia & Schwartz, 2002). The output should be read as a clearly-labeled conditional risk benchmark, not as an arbitrage-free market price.

## **4.2 Calibration and results**

Solar is used because public resource and production data are actually accessible — NASA's POWER satellite data and the PVWatts production model provide useful context for screening purposes, though an operational, revenue-grade claim would still need real meter or inverter records (NASA POWER, n.d.; National Laboratory of the Rockies, n.d.; Dobos, 2014; Sengupta et al., 2024). Taiwan is the one location with a fully preserved, disclosed volatility calculation: a four-day rolling smoothing pass plus a 1% outlier trim applied to the raw irradiance return data, producing an annualized volatility of about 189.5% from 2,166 transformed daily observations. The other four locations in this paper use specified scenario inputs rather than a preserved calibration trail, because their original calibration requests and diagnostics were not archived — a limitation this paper states plainly rather than papering over.

The Taiwan base case sets both the reference price and the strike at \$0.0525/kWh, a 3-month horizon, and a 2.5% discount-rate assumption. The binomial-tree price comes out to \$0.01917/kWh; the Monte Carlo price comes out to \$0.01957/kWh — about a 2.1% difference between the two independent methods, which is a good sign: it means the two calculations agree closely, and the result is stable once the tree has enough steps (roughly 400 or more). That agreement confirms the two pricing methods were implemented consistently with each other — it does not, on its own, confirm that the underlying assumptions (the volatility figure, the discount rate) are the economically correct ones; that still requires real-world validation this paper does not claim to have done.

The five-location comparison across Germany, Taiwan, Saudi Arabia, Arizona, and Brazil is a sensitivity exercise, not a ranking. Brazil and Taiwan produce larger option values than Germany simply because their specified price and volatility inputs are higher in this scenario set — the table

shows how much hidden parameter choices can move the size of the financial safeguard attached to a claim, not which location is objectively "riskier."

| Location     | Reference price<br>(\$/kWh) | Annual<br>volatility | Binomial price | Monte Carlo<br>price | Difference |
|--------------|-----------------------------|----------------------|----------------|----------------------|------------|
| Germany      | 0.0250                      | 45%                  | \$0.00234      | \$0.00236            | 0.9%       |
| Taiwan       | 0.0525                      | 189%                 | \$0.01917      | \$0.01957            | 2.1%       |
| Saudi Arabia | 0.0550                      | 172%                 | \$0.01841      | \$0.01876            | 1.9%       |
| Arizona, USA | 0.0580                      | 165%                 | \$0.01877      | \$0.01911            | 1.8%       |
| Brazil       | 0.0950                      | 198%                 | \$0.03702      | \$0.03781            | 2.1%       |

*Table 4. Cross-location option benchmark under stated inputs. Taiwan's volatility is a fully preserved, disclosed calculation; the other four locations are scenario inputs, not preserved calibrations.*

The same logic converts directly into margin and data-tolerance rules. A stated rule of 1.5 times the 99th-percentile daily loss produces about \$0.63/kWh of initial margin in the Taiwan base case. A tolerance rule allowing up to about a 21.7% measurement error, under a stated 95% variance-reduction target, sets a concrete bound on how much sensor or oracle error the system can absorb before it should stop trusting the data. These are policy-generated numbers, not observed market requirements — their value is that a later policy can use them directly to set an allowed quantity, a required margin, an admission bar, or a settlement safeguard, instead of picking one out of thin air.

One boundary governs how to interpret all of this: the underlying reference isn't actually traded, so this model supplies no arbitrage-free market evidence. Public resource data provides no physical settlement record on its own. Liquidity, basis risk, enforceability, real consumer demand, and physical delivery all remain outside what this pricing exercise establishes. Its contribution is a transparent path from modeled energy context to numbers a real policy can stress-test — not a claim to have found a universal energy price.

## 5. A Working, Tested Rule System

### 5.1 From five conditions to three decisions

The empirical result and the pricing result both point toward a simpler operational structure. Every energy-linked claim really only involves three distinct decisions: may the claim exist at all, how large may it be, and can it actually be settled later? Weak evidence, a binding size limit, and a later

settlement shortfall are three different kinds of failure — collapsing them into one overall score would hide exactly which part of the process actually failed.

The system built for this paper represents that sequence directly: a case and its evidence, a declared policy, an admission step, a set of quantity ceilings, a decision result, a later settlement outcome, and finally a receipt that preserves everything that went into the decision. Evidence and modeling context are recorded separately from each other. Admission returns a plain PASS or BLOCK. Quantity is capped by whichever applicable limit is lowest, expressed in the same units as the claim itself. Settlement is evaluated later, against whatever capacity is actually available at that time — not assumed automatically just because admission passed. A receipt preserves the inputs, the rule set used, whichever limit ended up binding, the final outcome, and which version of the rules and data were in effect.

This separation also clarifies what the software itself is actually responsible for. Smart contracts and deterministic code do not decide whether an underlying economic claim is sound — that judgment still belongs to the evidence and the policy. What the code can be properly evaluated on is whether it correctly preserves the separation between evidence, admission, quantity, settlement, and governance, rather than on whether a token transfer merely occurred.

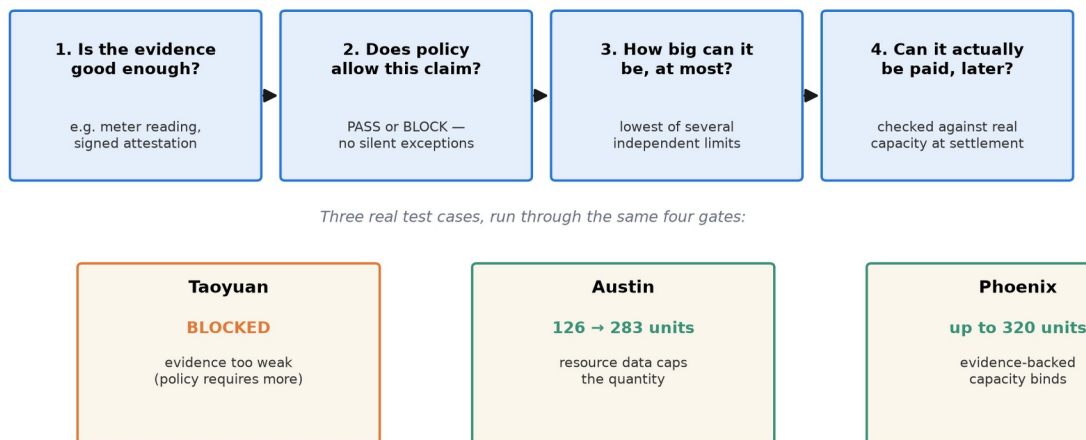


Figure 2. The four-gate decision process, applied to three real test cases with three different outcomes.

## 5.2 Running it on a real test network

The first layer of proof-of-concept testing runs selected rules after evidence has already been accepted. A meter-style reading is submitted; an accepted reading produces a fixed source hash; and a signed attestation ties together the recipient, the claimed surplus quantity, the measurement

window, a validity window, the source hash, the blockchain ID, and the contract address. Source and attestation hashes are consumed once used, which prevents the same piece of evidence from being reused to support more than one issuance.

An earlier proof run executed this exact path on Ethereum's Sepolia test network with a signed bundle covering 2,606.7 kWh of accepted surplus, minting 130.1697 SPK tokens at a stated \$0.05/kWh conversion rate. A later version of the system (SPK v1) shifted the default unit toward direct energy-native circulation — 1 kWh to 1 SPK — and left the dollar peg disabled entirely, to avoid quietly implying a fixed-price promise the system doesn't actually make. Invoice hashes prevent the same payment from being replayed twice. Redemption accounting converts spent tokens into owed-kWh claims that can later be marked fulfilled, in shortfall, or disputed.

What this establishes is bounded but real: accepted and rejected evidence paths can be kept genuinely separate, the same piece of evidence cannot be reused to support two different claims, payments carry replay protection, and redemption outcomes can actually be recorded. What it does not establish — and this paper says so directly — is revenue-grade metering, legally enforceable delivery, real-world reserve sufficiency, actual market demand, legal or tax classification, or production-grade cybersecurity. Those are conditions that belong to a future deployment decision, not to what this particular proof demonstrates.

### **5.3 Three real test cases**

The second testing layer checks the prior decision itself: why is a given claim admitted, what limits its size, and does the explanation stay consistent when the same policy is re-run? The test pack contains three fixture cases — Taoyuan, Austin, and Phoenix — each using model-based production data (PVWatts, plus typical-meteorological-year context) with the provider's identity, coordinates, system size, annual output, data source, and a cryptographic hash all preserved together.

Taoyuan gives the clearest test of the admission step. The pilot policy requires a minimum evidence-assurance level; the initial case as submitted only meets a lower level, so it is correctly BLOCKED, and no quantity is even calculated — there's nothing to size until the evidence bar is cleared. In a deliberate counterfactual test, the underlying evidence file and its hash are left completely unchanged, and only the declared assurance level is raised to meet the policy's bar. Admission then passes, the maximum allowed quantity comes out to 126 claim units, and the binding constraint recorded is specifically the provenance-policy rule — not some quiet change to the underlying evidence. This is the point of the test: the decision can be traced to exactly which rule changed, not treated as an unexplained number that moved.



Austin and Phoenix show that an admitted claim can still be limited for two entirely different reasons. Austin is admitted at 283.09811 units because its resource-context data — the modeled production evidence itself — is the lowest applicable ceiling. Phoenix is admitted at up to 320 units because a different limit, tied to evidence-backed capacity, is what binds there instead. Neither result says one location is objectively "better" than the other — it shows that each decision can be traced back to the specific governing policy and input that actually produced it, rather than presented as a black-box output.

Settlement is a separate result again. Running the Taoyuan 126-unit claim through three different settlement-capacity scenarios — 100%, 40%, and 0% of the claim actually available — correctly produces SETTLED, PARTIAL, and SHORTFALL outcomes respectively. At 40% capacity, the system records 50.4 units as covered and 75.6 units as an honest shortfall, rather than either overstating what was actually paid or silently rounding the gap away. Admission and quantity discipline reduce one category of risk; they do not, by themselves, guarantee that later performance will actually happen — and the system is built to say so explicitly rather than implying otherwise.

| Case / policy state                       | Decision         | Maximum allowed | What actually bound the decision          |
|-------------------------------------------|------------------|-----------------|-------------------------------------------|
| Taoyuan / lower assurance / pilot policy  | BLOCKED          | Not calculated  | Evidence assurance level (MIN_PROVENANCE) |
| Taoyuan / higher assurance / pilot policy | ADMIT WITH LIMIT | 126 units       | Provenance-policy rule                    |
| Austin / higher assurance / pilot policy  | ADMIT WITH LIMIT | 283.09811 units | Resource-context data ceiling             |
| Phoenix / open policy                     | ADMIT WITH LIMIT | 320 units       | Evidence-backed capacity                  |

*Table 5. Real, deterministic outcomes from the controlled test pack. The Taoyuan counterfactual changes only the declared assurance state — the evidence file and its hash are untouched.*

## 5.4 Keeping a record of how each decision was actually made

Every decision produces a receipt that preserves more than just the final outcome — it records the case and decision identifiers, the policy version and its hash, the evidence hashes involved, the specific context used, which rules were actually evaluated, whichever limit ended up binding, the final result, and which data revision was in effect. Provenance is treated as part of the result itself: a

reviewer can inspect not just what the system decided, but exactly how that decision was produced (Moreau & Missier, 2013).

The version of the system reviewed for this paper was frozen at one specific, recorded source revision. At that exact revision, the deterministic core passed 60 out of 60 tests, and the front-end passed 64 out of 64 tests; the production build, browser walkthrough, continuous-integration pipeline, smart-contract checks, security checks, and secrets checks all passed as well. This supports the claim that the implementation was internally consistent and correctly built at the point it was reviewed. It is not a substitute for independent operator validation or real-world commercial testing — those remain separate, unresolved questions.

The resulting contribution is deliberately narrower, and more useful, than a product pitch. Under stated, controlled assumptions, the five-condition framework can be represented as an auditable decision-and-settlement process. It correctly exposes weak evidence, correctly identifies whichever limit is actually lowest and binding, keeps settlement genuinely separate from admission, and preserves the full evidence-and-policy trail behind every decision it makes.

## **6. Discussion**

### **6.1 What this evidence actually changes**

The paper starts from a common, intuitive belief: real energy expenditure or real energy production should make a digital claim more credible. The evidence shows that intuition is incomplete, not wrong. Bitcoin's cumulative mining expenditure is real, costly, and observable — but the CEIR association it produces isn't unique to electricity prices, and it doesn't grant holders any redemption right regardless. Renewable output is genuinely productive — but its financial value depends on location, timing, uncertainty, and settlement terms that a bare energy reference doesn't specify. Code can enforce a rule with perfect technical precision — but it cannot, on its own, establish that the rule it's enforcing is economically sound.

The right unit to analyze, in other words, is the whole claim process — not just the energy narrative attached to it. A physical reference answers what a claim points to. A financial constraint answers what the evidence actually permits, how uncertainty changes the allowed quantity or the required protection, what must happen at settlement, and who is allowed to revise any of those rules later. Energy can only contribute real credibility through that full institutional chain — not by itself.

This reading deliberately avoids two opposite mistakes. It rejects the idea that proof-of-work spending automatically provides intrinsic backing — the empirical section shows plainly that it

doesn't, at least not through this mechanism. It also rejects the opposite over-correction, that energy references are therefore financially meaningless. Verified production evidence genuinely can limit a digital claim — but only once that evidentiary state is actually tied to rule-bound admission, compatible quantity ceilings, real settlement accounting, and limited governance, all four together.

## **6.2 What this means for digital finance going forward**

For token design: a token doesn't become credible just because some underlying asset happens to be real. The mapping from evidence to claim has to be explicitly defined. Modeled solar potential, an actual meter reading, exported electricity, contracted value, and final physical delivery should not be treated as interchangeable numbers — collapsing those distinct states into one is exactly what invites over-issuance and makes later disputes hard to resolve fairly.

For pricing: a financial model shouldn't be judged purely on how sophisticated its math looks. A model becomes institutionally useful once a pre-specified policy actually maps its output into a real limit or safeguard. This paper's cold-start benchmark shows concretely how price and volatility assumptions move option value, margin, and data tolerance — a future pilot would still need to validate those assumptions against real outcomes, and specify in advance exactly how a result changes an allowed quantity or a collateral requirement.

For governance: energy cannot constrain a system if the people running it can mint without accepted evidence, quietly revise a pricing rule after seeing an inconvenient result, or waive a settlement obligation whenever it becomes costly to honor. Who has the authority to change the rules is part of the financial claim itself — not a separate, purely technical software-management concern.

For research design: a negative empirical result can actually sharpen constructive work, once it's used to retire an unsupported shortcut rather than defended past the point it holds up. The CEIR result that failed here narrows the architecture in a useful way — Bitcoin is not carried forward into this paper's claim as a valuation foundation. Instead, the empirical chapter's honest failure is exactly what identifies which rules the later, constructive sections need to specify explicitly, rather than assume.

## **6.3 What this paper does not establish, and how it could be proven wrong**

The empirical evidence here is limited to one proof-of-work asset, and to a market-capitalization series derived from a locally reconstructed price export and an approximate supply curve, not a fully documented observed series. Mining electricity use is itself modeled rather than metered, the intended geography-weighted price series failed to merge correctly, and the ban-date split used is a reasonable researcher choice rather than an official event date. These are real defects, and they're

central to why the identification failed — but they also mean this paper hasn't yet delivered a clean test of a properly constructed, geography-sensitive mining-cost measure. That remains open.

The pricing layer is still a cold-start scenario model. Real power markets can show sudden jumps, negative prices, mean reversion, seasonal patterns, curtailment, basis risk, counterparty risk, and regulatory constraints that go beyond the simple diffusion model used here. Only Taiwan has a fully preserved, disclosed volatility calculation — the other four locations are explicitly illustrative scenarios, not calibrated results, and this paper does not claim otherwise.

The implementation is controlled proof-of-concept evidence, not a finished product. Public test-network execution cannot establish revenue-grade metering, legal enforceability, real demand, tax treatment, regulatory compliance, or physical redemption. The test cases use fixture data, and the current policy pack does not yet convert the pricing benchmark into a validated, real-world risk-based quantity rule.

These limitations are exactly what would make the architecture falsifiable, if someone wanted to test it properly: the framework would be genuinely weakened if real operator evidence turned out to resist consistent classification, if pricing outputs couldn't be converted into defensible quantity limits, if the same policy produced unpredictably different admission outcomes under equivalent evidence, if settlement rights stayed ambiguous in practice, or if governance could waive the constraint whenever it became inconvenient to honor. A future pilot would need to record the operator, the data source, the measurement window, the evidence-assurance state, the policy version, the risk rule, the admission result, the quantity limit, the settlement outcome, and the receipt — all before observing how the claim actually performs, not after.

## **7. Conclusion**

This paper asks a specific question: when can energy actually discipline a digital financial claim? The honest answer is: only conditionally. Expenditure and production genuinely provide a physical reference point, but it's evidence, issuance rules, risk controls, settlement obligations, and limited governance — held jointly — that determine whether that reference actually limits the claim in practice, rather than just describing it.

The Bitcoin analysis reproduces an attractive-looking CEIR relationship, and then shows carefully why it does not actually identify an electricity-price anchor — the same pattern shows up just as strongly using cumulative electricity use, or even the simple passage of time. The pricing benchmark converts renewable-output uncertainty into inspectable financial numbers, without dressing them up as observed market prices. The working rule system keeps admission, quantity, and settlement as

genuinely separate decisions, and preserves the evidence and policy trail behind each one — and it's been tested, not just proposed, on real cases with three different real outcomes.

The resulting standard is simple to state, even though building it is not: an energy-linked claim shouldn't exist unless its evidence, its governing policy, its quantity limit, its settlement obligation, and the record of how it was decided can all actually be inspected and enforced. Energy becomes genuinely credible not when a system merely says that energy was spent or produced, but when the system demonstrably cannot create or settle claims beyond what its own evidence and rules can actually support.

## **Data, Code, and Artifact Statement**

The empirical analysis uses a locally reconstructed Bitcoin price export, an approximate supply curve, Cambridge Bitcoin Electricity Consumption Index exports, Cambridge Mining Map context, and the Alternative.me Fear & Greed Index. The audited manuscript records source-lineage limitations directly, including the missing upstream price-export metadata and the failed geography-weighted electricity-price merge.

The pricing analysis uses archived scripts and output files for the Taiwan volatility transformation, the binomial and Monte Carlo valuations, the cross-location scenarios, margin stress testing, and oracle tolerance. The implementation evidence is documented in the SolarPunk project artifacts, including Sepolia test-network runtime records, the deterministic case and policy manifests, receipts, controlled test outcomes, and validation logs. The exact frozen case-workbench revision is recorded as: eb8714a6544b3480226283a69d41b3946df63451

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